

Question	Answer	Mark
6(a)	$R = \rho L / A$	C1
	$3.0 = \rho / [\pi \times (0.48 \times 10^{-3} / 2)^2]$	C1
	$\rho = 5.4 \times 10^{-7} \Omega \text{m}$	A1
6(b)(i)	p.d. = $5.0 - (2.0 \times 1.6)$ $= 1.8 \text{V}$	A1
6(b)(ii)1.	current in resistor = $1.8 / 4.5 (= 0.40 \text{A})$	C1
	current in wire = $1.6 - 0.40 (= 1.2 \text{A})$	C1
	$R_X = 1.8 / 1.2$ $= 1.5 \Omega$	A1
	or	
	$R_T = 1.8 / 1.6$ or $(5.0 / 1.6) - 2.0 (= 1.125 \Omega)$	(C1)
	$(1 / 1.125) = (1 / 4.5) + (1 / R_X)$	(C1)
	$R_X = 1.5 \Omega$	(A1)
6(b)(ii)2.	length = $1.5 / 3.0$ or $1.5 \times 1.8 \times 10^{-7} / (5.4 \times 10^{-7})$ $= 0.50 \text{m}$	A1

9702/23

Cambridge International AS/A Level – Mark Scheme

May/June 2018

PUBLISHED

Question	Answer	Mark
3(c)(i)	Q plotted at (82, 240)	A1